

CompTia A+ Assessment

Section A: Concept Application

Scenario 1:

Selecting Compatible PC Components

1. Choose the CPU first:

- Select a CPU suitable for office work such as documents, web browsing, and video calls.
- Note its socket type and supported RAM type.

2. Choose a compatible motherboard

- The motherboard must have the same CPU socket as the CPU.
- Check that it supports the required RAM type, such as DDR4 or DDR5.

3. Choose the RAM

- Select RAM supported by the motherboard and CPU.
- Check the DDR generation, speed, capacity, and number of slots.

4.Choose the storage

- Select an SSD for faster booting and application loading.
- Check whether the motherboard supports the chosen M.2 NVMe or SATA SSD.

5.Confirm compatibility

- Check the motherboard manufacturer's CPU support list and specifications.
- Compare the CPU socket, RAM type/speed, and storage interface with the motherboard specifications.

Scenario (2):

	HDD	SDD	NVMe
Technology	Magnetic disks	Flash memory + SATA	Flash memory + PCIe
Speed	Slow	Fast	Very Fast
Cost	Low	Medium	Higher
Best Use	Large file storage	General office use	Fast applications and heavy workloads

Recommendation: NVMe SSD

- For a modern office workstation, I would recommend an NVMe SSD because it provides much faster data transfer and lower latency than a traditional HDD or SATA SSD. It can make Windows boot faster, applications open quickly, and file transfers more responsive.

- Technical reason: NVMe communicates directly through the PCIe interface, while a SATA SSD is limited by the SATA interface. HDDs are slower because they use mechanical moving parts.
- Example: A 500 GB or 1 TB NVMe SSD would be a good choice for an office workstation.

Scenario (3):

Role of GPU and Cooling System:

- GPU (Graphics Processing Unit):
- Handles graphics and video processing.
- Helps run games, videos, and graphics-heavy applications smoothly.
- Under sustained load, the GPU produces more heat and uses more power.
- A good GPU maintains stable performance without frequent crashes or slowdowns.

Cooling System:

- Removes heat from the CPU and GPU.
- Fans and heatsinks move heat away from important components.

- Good airflow prevents overheating and thermal throttling.
- Clean fans and properly applied thermal paste help maintain stable temperatures.

The GPU provides the required graphics performance, while the cooling system keeps the GPU and other components at safe temperatures. Together, they help the PC maintain stable performance during long periods of heavy use.

Scenario (4):

Why Hardware Compatibility Matters:

Hardware compatibility is important because all computer components must work together correctly. Before assembling a PC, we should check that the CPU, motherboard, RAM, GPU, power supply, and storage are compatible.

Problems from Incompatible Components:

- **CPU + Motherboard:** Wrong CPU socket → PC may not start.

- **RAM + Motherboard:** Wrong DDR type → RAM will not work.
- **GPU + Motherboard/Power Supply:** Not enough space or power → GPU may not work properly.
- **Storage + Motherboard:** Unsupported M.2/SATA interface → storage may not be detected.
- **Power Supply:** Insufficient wattage → crashes, shutdowns, or unstable performance.
- **Cooling:** Incompatible or insufficient cooler → CPU may overheat.

Scenario 5:

Choosing a PSU:

1. Calculate power needs:

- Check the power consumption of the CPU, GPU, storage, fans, and other components.

2. Add extra capacity:

- Choose a PSU with around 20–30% more wattage than the estimated maximum load.

3. Check PSU quality:

- Choose a reliable PSU with an 80 PLUS efficiency rating and suitable protections.

4. Check connectors:

- Make sure it has the required 24-pin motherboard, CPU 4/8-pin, GPU PCIe, and SATA power connectors.

Safety and Connection Checks:

1. Turn off and unplug the PSU before connecting components.
2. Make sure the 24-pin motherboard cable is fully connected.
3. Connect the CPU power cable near the processor.
4. Connect the required GPU power cables.
5. Connect SATA power to HDDs/SSDs if needed.
6. Check that cables are firmly connected and not damaged.
7. Make sure no loose screws or metal objects are inside the case.
8. Confirm the PSU voltage switch, if present, is set correctly for the local supply.
9. Do not mix modular cables from different PSU models.

10. Finally, connect AC power and switch on the PC.

Scenario (6):

POST (Power-On Self-Test) Checks:

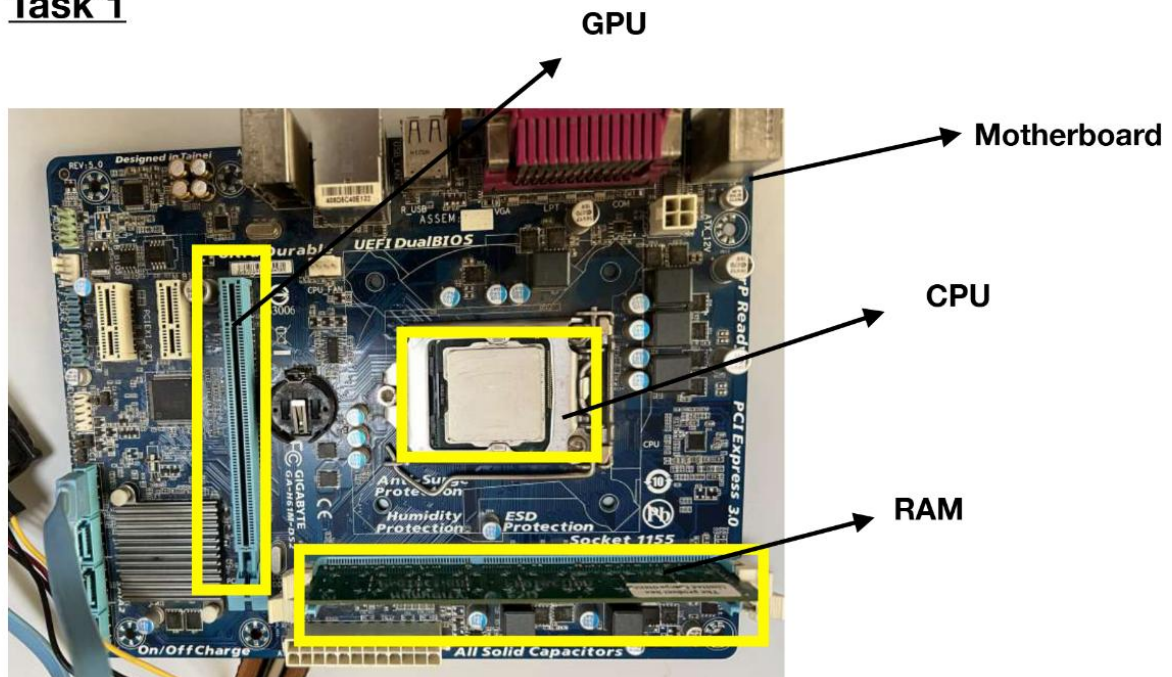
1. **Power on the PC:**
 - Check that fans spin and power LEDs turn on.
2. **Check the display:**
 - Confirm the BIOS/UEFI screen appears without errors.
3. **Listen for beep codes:**
 - Check for unusual beeps that indicate hardware problems.
4. **Check BIOS/UEFI:**
 - Confirm the CPU, RAM, and storage devices are detected correctly.
5. **Check for error messages:**
 - Make sure there are no POST warnings or hardware errors.

If POST Fails:

- Turn off the PC and check all power connections.
- Reseat the RAM and GPU.
- Test RAM modules one at a time.
- Check the CPU and motherboard connections.
- Disconnect unnecessary devices and try booting with minimum hardware.
- Use the motherboard's beep codes or diagnostic LEDs to identify the faulty component.

Section B: Practical Coding Tasks

Task 1



Cooling FAN



Storage Devices (SATA SSD)

Make/Model of each component:

1. Motherboard - Gigabyte GA-H61M-DS2
2. CPU - Intel, Socket LGA 1155
3. RAM - One DDR3 DIMM installed (single-channel)
4. GPU - Dedicated Card Slot
5. Storage - WD Green SATA SSD, 120GB
6. Cooling System - Intel stock cooler

Functions:

1. Motherboard:

- The main circuit board connecting and enabling communication between every other component — CPU, RAM, storage, GPU, and peripherals.

2. CPU:

- Executes instructions and performs calculations; acts as the primary processing unit ("brain") of the system.

3. RAM:

- Provides temporary, high-speed memory for data and programs actively in use - cleared when the system powers off.

4. GPU:

- Handles graphics rendering and parallel processing tasks - critical for display output and any graphics-intensive workloads.

5. Storage device:

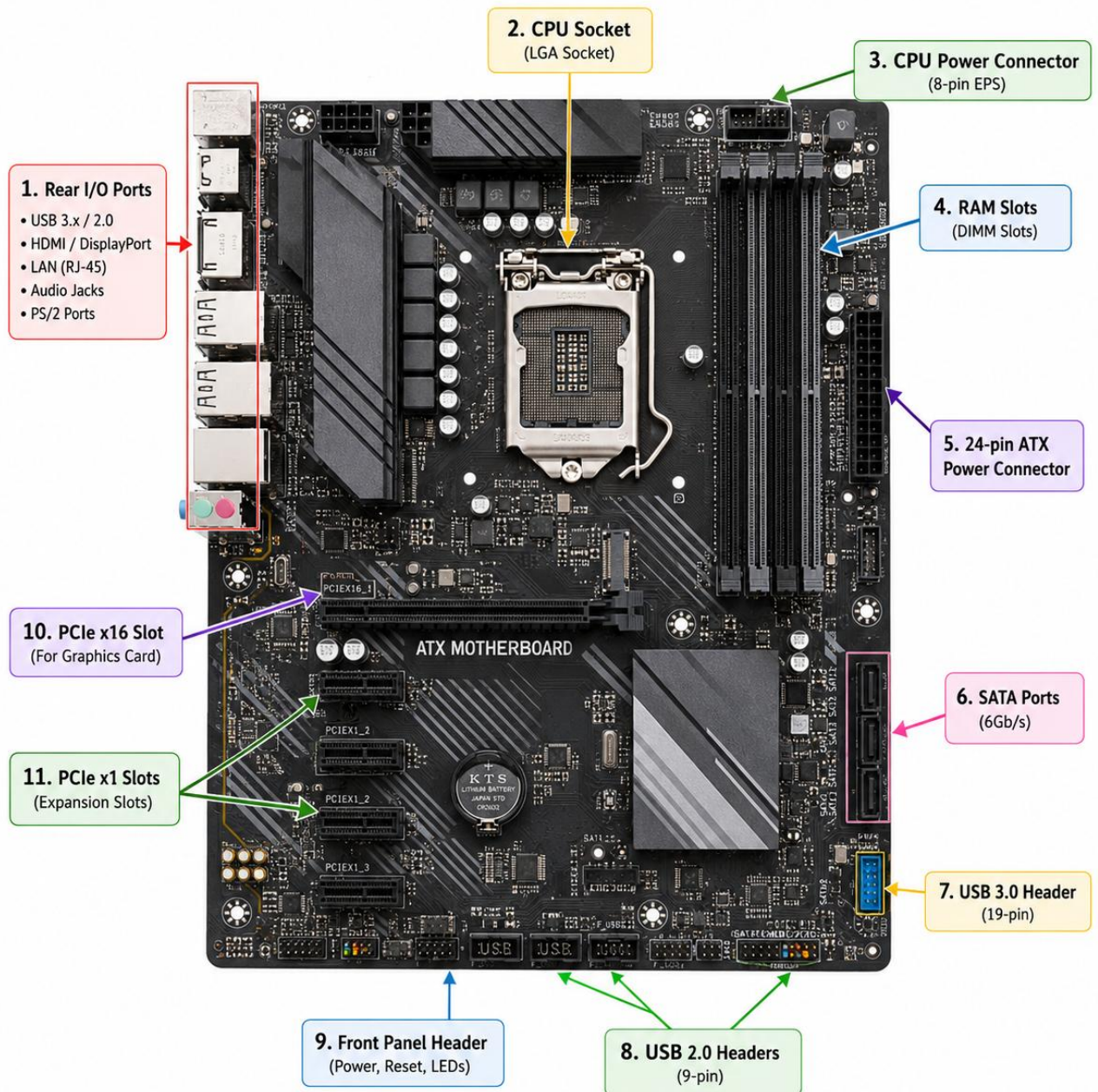
- Provides long-term, non-volatile storage for the operating system, applications, and files.

6. Cooling System:

- Dissipates heat generated by the CPU/GPU during operation, preventing thermal throttling and maintaining stable performance.

Task (2)

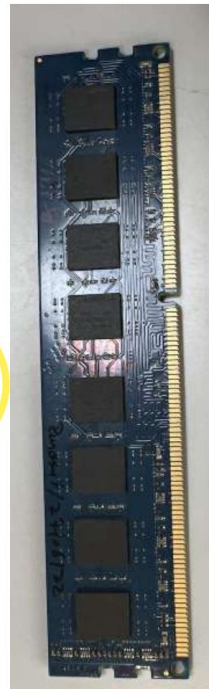
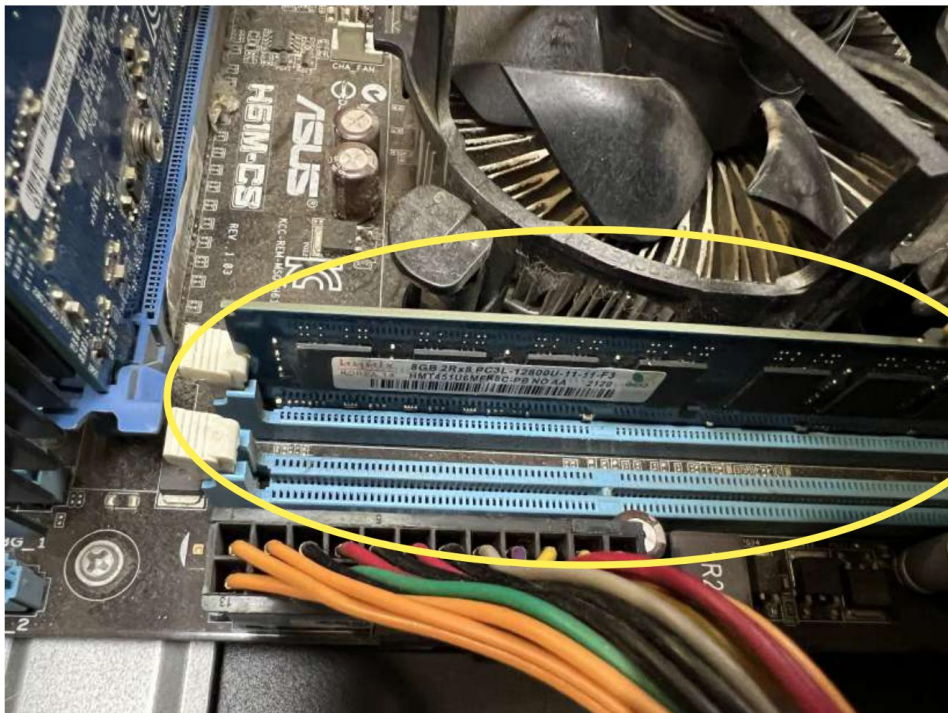
ATX MOTHERBOARD – Labeled Diagram



1. CPU Socket
2. RAM Slots (DIMM Slots)
3. PCIe x16 Slot

4. PCIe x1 Slots
5. SATA Ports
6. 24-pin ATX Power Connector
7. CPU Power Connector (8-pin EPS)
8. USB Headers (USB 2.0/3.0):
9. Front Panel Header
10. Rear I/O Ports

Task (3)



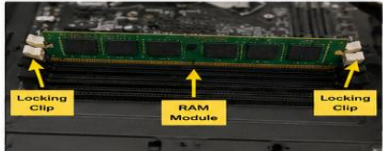
RAM Module



SATA SDD

Task (4):

Desktop PC Disassembly – Step by Step

Step 1: Unplug the Power  Turn off the computer and unplug the power cable. Disconnect all other external cables.	Step 2: Remove the Side Panel  Remove the thumb screws from the back of the cabinet and slide the side panel backward to take it off.	Step 3: Disconnect Cables  Disconnect power cables from the motherboard, GPU and drives. Remove SATA data cables from HDD/SSD.
Step 4: Remove the RAM  Press the locking clips on both sides of the RAM slot. The module will pop up. Gently lift it out.	Step 5: Remove HDD/SSD  Unscrew the drive from the drive bay. Disconnect the SATA data cable and power cable. Remove the drive.	Step 6: Remove Graphics Card (GPU)  Remove the retaining screw at the back. Press the PCIe slot latch and gently pull the graphics card out.
Step 7: Remove CPU Cooler  Unplug the CPU fan cable. Loosen the mounting screws (or push the clips) and carefully lift the cooler off the CPU.	Step 8: Remove Motherboard  Disconnect all remaining cables from the motherboard. Remove the mounting screws and carefully lift the motherboard out of the cabinet.	Step 9: Organize Components  Keep all components and screws in order. Store screws in a small box so nothing gets lost.

➤ **Step 1: Unplug the Power**

- Turn off the computer and unplug the power cable.
- Disconnect all external devices such as the monitor, keyboard, mouse, and network cable.

➤ **Step 2: Remove the Side Panel**

- Unscrew the side panel of the PC cabinet.
- Slide the panel backward and remove it carefully to access the internal components.

➤ **Step 3: Disconnect Cables**

- Disconnect the 24-pin motherboard power cable, CPU power cable, SATA cables, and GPU power cable.
- Keep the cables organized for easy reassembly.

➤ **Step 4: Remove the RAM**

- Press the locking clips on both sides of the RAM slot.
- Gently lift the RAM module out of the motherboard.

➤ Step 5: Remove the HDD/SSD

- Unscrew the HDD/SSD from its mounting bracket.
- Disconnect the SATA data cable and power cable, then remove the drive.

➤ Step 6: Remove the Graphics Card (GPU)

- Unscrew the graphics card from the case.
- Press the PCIe slot latch and carefully pull the GPU out.

➤ Step 7: Remove the CPU Cooler

- Disconnect the CPU fan cable.
- Unscrew or unclip the cooler and carefully lift it off the CPU.

➤ Step 8: Remove the Motherboard

- Disconnect any remaining cables.
- Remove the motherboard mounting screws and carefully lift the motherboard out of the cabinet.

➤ **Step 9: Organize the Components**

- **Arrange all removed components neatly.**
- **Store screws in a small container to prevent them from getting lost.**

SECTION C: Mini Capstone Project

Three components to check first on boot failure:

- **RAM** (most common cause — reseal/try single stick), power connections (24-pin + CPU power fully seated), and the GPU (reseated in slot, PCIe power connected).

Motherboard–CPU compatibility:

- **CPU socket:** The CPU socket must match the motherboard socket.
- **Chipset:** The motherboard chipset should support the CPU.
- **BIOS:** The motherboard BIOS may need an update to support the CPU.
- **Power:** The motherboard must provide enough power for the CPU.

Verifying RAM and storage installation:

- **RAM:** Check that the RAM is fully inserted into the correct slot and both side clips are locked.
- **Storage:** Check that the SSD or HDD is securely installed and its data and power cables are connected properly.
- **Test:** Turn on the PC and check the BIOS/UEFI to make sure the RAM and storage device are detected.

Hardware-installation checklist (3+ steps)

1. **Install the CPU:** Place the CPU correctly in the motherboard socket.
2. **Install the RAM:** Insert the RAM into the correct slots until it clicks.
3. **Install the storage:** Connect the SSD/HDD properly.
4. **Install the GPU:** Insert the GPU into the PCIe slot and secure it.

5. **Connect the PSU cables:** Connect power cables to the motherboard, GPU, and storage.
6. **Check all connections:** Make sure everything is secure before turning on the PC.

SSD vs HDD justification for the lab

- SSD is the **Faster** and HDD is the **Slower**.
- SSD is **no moving parts** and HDD **has moving parts**.
- SSD **uses less power** and HDD **uses more power**.
- SSD is **more expensive** and HDD is **cheaper**.

SECTION D: AI-Augmented Learning: Approach

1. The exact prompt I gave the AI tool.

- Guide a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly. Explain the troubleshooting steps in the correct priority order, starting with basic power checks and then checking the motherboard, CPU, RAM, GPU, storage, and cables. Use simple and easy-to-understand words.

2. The AI's original checklist and your corrected version.

AI's Original Checklist

1. Check if the PC power cable is connected.
2. Check the power supply (PSU).
3. Check the RAM.
4. Check the CPU.
5. Check the GPU.
6. Check the storage device.
7. Check all cables and connections.
8. Try turning on the PC again.

Corrected Version

1. **Check power first:** Make sure the power cable, PSU switch, and wall socket are working.
 2. **Check motherboard power cables:** Confirm the 24-pin motherboard cable and CPU power cable are connected.
 3. **Check RAM:** Remove and reinstall the RAM firmly in the correct slot.
 4. **Check CPU and cooler:** Make sure the CPU is installed correctly and the cooler is connected.
 5. **Check GPU:** Make sure the GPU is properly seated and its power cable is connected if required.
 6. **Check storage:** Confirm the SSD/HDD is connected correctly.
 7. **Check POST/BIOS:** Look for beep codes, warning LEDs, or display messages.
 8. **Test again:** Power on the PC and confirm that it reaches the BIOS or operating system.
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